Grouped Subspace Linear Semantic Alignment for Hyperspectral Image Transfer Learning

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***Abstract*— Transfer learning (TL) offers an effective way to reduce the demand for labeled samples in remote sensing image classification. However, existing TL methods have some limitations. Simple linear TL methods cannot align the source and TARget domains (TARs) well when the data shift is complicated, while nonlinear methods consist of many learnable parameters and often need many labeled samples. To overcome these issues, we design a novel grouped subspace linear semantic alignment (G-SLSA) algorithm, which consists of four main ingredients. First, inspired by the linear supervised transfer learning (LSTL) approach, we propose subspace linear semantic alignment (SLSA) aiming to reduce the demand for labeled samples. Second, considering the heterogeneity of class-level data shift, we extend SLSA to G-SLSA through a grouped alignment strategy, which can reduce the data shift by decomposing a multiclass TL task into a set of binary subtasks. Third, considering the demand of subtasks fusion on posterior probabilities, we propose a robust posterior probability estimation method for the binary generalized learning vector quantization (GLVQ) that is used in G-SLSA. Finally, a pairwise coupling (PWC) method is applied to fuse the results of each subtask. Experimental results conducted on three popular hyperspectral datasets demonstrate that G-SLSA outperforms other traditional and state-of-the-art deep learning (DL) methods, with an overall accuracy (OA) of 80.78±4.28% for Pavia City-University of Pavia (PC-UP) dataset TL scenario when five samples per class are available in the TAR.**

***Index Terms*— Generalized learning vector quantization (GLVQ), hyperspectral image (HSI) classification, subspace alignment (SA), transfer learning (TL).**

I. Introduction

HYPERSPECTRAL images (HSIs), which can provide abundant information for the accurate discrimination of similar materials of interest, consist of data gathered in hundreds of spectral bands. HSI classification has been widely applied in many fields, such as precision agriculture, forestry monitoring, mineral exploitation, and environmental assessment [1]–[4].

Advanced supervised learning methods [5]–[10] have greatly facilitated the development of HSI classification. However, these methods heavily rely on a large number of labeled samples, which limits their breadth of application and usage [11], [12]. In the last decade, transfer learning (TL) has been extensively studied with the aim of either reducing the demand for labeling or improving performance by learning from different but related tasks [13]–[17]. Concretely, in remote sensing applications, TL uses knowledge for classification from previously labeled images (source domain) to newly acquired images [TARget domain (TAR)] or from areas that are more easily labeled (source domain) to areas that are more difficult to label (TAR). We usually differentiate two different settings, including unsupervised and supervised TL [18]. In this article, we mainly focus on the supervised case, and the proposed method is conducted on the shared classes between different domains.

The unsupervised setting is also referred to as unsupervised domain adaptation (UDA), which transfer source-domain knowledge learned from labeled source data to TAR with unlabeled target data only. A common UDA strategy is to find a domain-invariant feature space for the source and TARs [19]. In [20], Fernando *et al.* sought a domain-invariant feature representation by learning a rotation transformation that aligns the principal subspace (spanned by the d principal components corresponding to the d largest eigenvalues) of the source domain with that of the TAR, which is known as subspace alignment (SA). Gong *et al.* [21] proposed a geodesic flow kernel with the aim of modeling incremental changes between the source and TARs and finding the best intermediate subspace on a geodesic path as the domain-invariant feature space. The objective of transfer component analysis is to estimate the distribution distance between two domains based on the maximum mean discrepancy and find a shared subspace spanned by transfer components in which the data distributions in the different domains are close to each other [22]. Considering both geometric alignment and statistical alignment, Zhang *et al.* [23] propose the discriminative cooperative alignment for HSI classification, in which the domain-invariant subspace is learned under the joint work of SA and distribution alignment. Although not sufficiently effective, UDA methods have still been successfully applied to text classification [24], [25] and object recognition [26], [27]. However, they are rarely directly applied to HSI classification. The reasons include the following: 1) the primary purpose of pixel-level HSI classification is thematic mapping, which has high requirements in terms of the accuracy of the data source, i.e., the classification results and 2) in addition, some UDA algorithms require

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very large-scale matrix operations for pixel-level classification. Therefore, in practice, it is extremely challenging to achieve sufficient classification accuracy without the aid of a few labeled target samples.

In the supervised setting, both source- and limited target-domain labeled data are available. Deep TL is widely known to be the most successful method in this scenario [28]. However, the number of learnable parameters in a deep learning (DL) model is much larger than that in traditional machine learning (ML) methods, which will lead to harmful model overfitting [29]. Recent techniques, such as the episodic training scheme [30] and the Siamese neural network (SNN) [31], have suggested that training deep neural networks (DNNs) with various combinations of limited labeled samples can address model overfitting. Episodic training schemes utilize sampled minibatches called episodes during training, which are made by subsampling classes as well as samples to mimic a few-shot task by subsampling training set to endow a trained model with better generalization performance [32]. An SNN consists of twin networks that accept distinct inputs but are joined by a final cost function [33], making twin networks can be trained with pairs of samples drawn from different domains to narrow the data shift [34]. Recent studies applied these approaches in HSI classification [29], [35]–[37].

A traditional ML model usually has fewer parameters than a DL model. Therefore, TL methods based on traditional ML models are less likely to suffer from overfitting. An example of this is adaptive support vector machines (A-SVMs) [38], which realize the adaptation of a source model to a TAR by learning a new decision boundary from the source model and target training samples. The new decision boundary is close to the boundary of the source model but can still separate the target training samples well. However, this method is highly sensitive to an empirical parameter, i.e., the cost factor, which balances the contribution between the source classifier and training samples. NFNalign is a parameter-robust hyperspectral domain adaptation method based on nonlinear feature normalization (NFN) [39]. NFN can align the hyperspectral data with reference spectra B to produce a linearized representation. Therefore, for domain adaptation, NFNalign transforms hyperspectral data in different domains to a common domain $\mathcal{C}$ by NFN with the same B. Although NFNalign is robust toward parameter variations, the determination of the reference spectra B affects the algorithm's performance significantly. Another method of note is linear supervised transfer learning (LSTL) [40]. LSTL achieves adaptation of the source model by learning a linear mapping to project the target data into the source domain. The learning problem can be formulated as an optimization problem such that the optimal linear mapping minimizes the classification loss for classifying the projected target-domain labeled data using a source-domain generalized learning vector quantization (GLVQ) model [41], which is a powerful prototype-based classifier. Thus, the source-domain GLVQ model can generate correct predictions for the transformed target-domain data, as the learned linear mapping can reduce the data shift. Similarly, LSTL involves fewer learnable parameters, enabling it to work with limited label information. However, we argue that LSTL suffers from several major drawbacks: 1) it is a global alignment strategy and, as such, does not consider that the data shift may be different between classes, which may result in a locally negative transfer [42]; 2) it will ignore the translational component of the data shift because of the lack of constant terms; and 3) it may encounter difficulties with high-dimensional data, such as HSIs, because an excessively high dimensionality leads to a larger and less accurate mapping matrix. To overcome these drawbacks, this work makes the following contributions.

- We propose subspace linear semantic alignment (SLSA) by combining LSTL and SA, and we introduce constant terms into the learnable linear transformation, thus significantly improving the alignment performance.
- We propose a grouped alignment strategy that integrates SLSA, presenting the grouped subspace linear semantic alignment (G-SLSA) algorithm for HSI TL. The grouped alignment strategy can reduce the data shift and alleviate the negative transfer problem of global alignment.
- We present a posterior probability estimation method for binary GLVQ. Therefore, instead of voting, G-SLSA can fuse the binary subtasks via a pairwise coupling (PWC) method [43], thereby obtaining better fusion results.

II. Methodology

As shown in the flowchart in Fig. 1, the proposed method mainly includes four steps: 1) first, we adopt the grouped alignment strategy to decompose the K-classes TL task into $K(K-1)/2$ binary subtasks; 2) next, for each subtask, we perform the SLSA algorithm separately, thereby obtaining pairwise GLVQ models for the TAR; 3) then, we obtain the GLVQ probabilistic output for each subtask; and 4) finally, we fuse all subtasks by PWC. The flowchart of the SLSA model is presented at the bottom of Fig. 1, which is the second step of G-SLSA and consists of a reverse version of SA followed by the improved LSTL to learn the linear transformation.

A. SA Using Reverse Mapping

The main idea of SA is to learn a rotation transformation to align the source-domain principle subspace with the target-domain principle subspace and build a new representation that is invariant between the two domains. A source classifier is trained after the transformation of source-domain data. To ensure the consistency of the transformation direction, we perform the alignment via reverse mapping, that is, aligning the target-domain subspace with the source-domain subspace. The corresponding pseudocode is presented in Algorithm 1.

In addition, for HSIs data, we recommend performing channelwise $0\sim1$ normalization (rescaling the range of the data to the interval [0, 1]) before principal component analysis (PCA) to obtain better results.

Similar to other UDA methods, SA can slightly improve the classification results but cannot achieve promising performance for HSI classification tasks. There are two possible reasons for this: First, SA involves only a rotation transformation and does not consider a scale transformation. Second, a lack of target-domain label information will cause SA and other classic UDA methods to fail to obtain satisfactory results.

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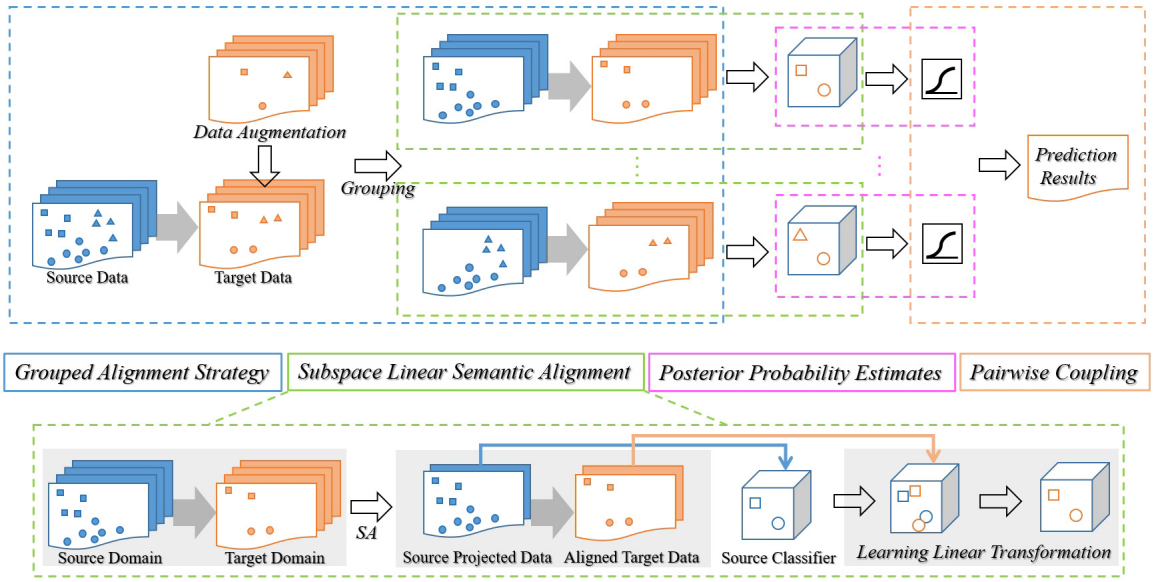


Fig. 1. Flowchart of the proposed method. (Top) Flowchart of G-SLSA, which includes four steps: 1) adopt the grouped alignment strategy to decompose the K-classes TL task into $K(K-1)/2$ binary subtasks; 2) perform the SLSA algorithm separately and obtain pairwise GLVQ models for the TAR; 3) then, obtain the probabilistic output for each subtask; and 4) fuse all subtasks by PWC. (Bottom) Flowchart of SLSA for binary classification, which is the second step of G-SLSA.

Algorithm 1 SA Using Reverse Mapping (Adapted From [20])

Data : Source data S, Target data T, Source labels L_s, Subspace dimensionality d.
Result : Predicted target labels L_t.
$X_s \leftarrow PCA(S,\ d)$;
$X_t \leftarrow PCA(T,\ d)$;
$X_a = X_t X_t^{\top} X_s$;
$S_s = SX_s$;
$T_a = TX_a$;
$L_t \leftarrow Classifier(S_s,\ T_a,\ L_s)$.

B. Subspace Linear Semantic Alignment (SLSA)

We argue that further data shift reduction can be achieved by introducing a scale transformation and target-domain label information when learning the linear mapping M. To achieve these two modifications, we take inspiration from the LSTL algorithm and try to perform SA followed by LSTL.

First, we perform Algorithm 1 on the source- and target-domain data and skip the classification step.

Second, a GLVQ model is trained with labeled data from the source domain to obtain the prototype vectors W and prototype labels L_w. The cost function of GLVQ is defined as

$$E_s = \sum_{i=1}^{N_s} \mu\left(S_i X_s - W_i^+, S_i X_s - W_i^-\right) \tag{1}$$

where N_s is the number of source-domain training samples, S_i is the ith labeled source-domain sample, W_i^+ is the closest prototype vector with the same label as S_i, and W_i^- is the closest prototype vector with a different label than S_i. The function μ is defined as

$$\mu(\delta^+, \delta^-) = \frac{\|\delta^+\|_2^2 - \|\delta^-\|_2^2}{\|\delta^+\|_2^2 + \|\delta^-\|_2^2} \tag{2}$$

where δ^+ and δ^- are two inputs of μ. For example, $\delta^+ = S_i X_s - W_i^+$ and $\delta^- = S_i X_s - W_i^-$ in (1).

Third, the GLVQ model is adapted to the TAR by learning a linear mapping H to map the target-domain data into the source domain. The cost function of GLVQ over target transformed labeled data is defined as

$$E_t = \sum_{i=1}^{N_t} \mu\left(T_i^l X_t M H - W_i^+, T_i^l X_t M H - W_i^-\right) \tag{3}$$

where N_t is the number of target-domain training samples.

Note that H is a nondiagonal matrix; therefore, we can perform rotation and scaling simultaneously with H and omit the transformation M. In addition, to improve the fitting performance of H, we introduce constant terms into it by representing the target-domain data in homogeneous coordinates. The aligned target-domain data are represented as $[TX_t, \mathbf{e}]H$, where $\mathbf{e}$ is a column vector of all ones, and the size of H has changed from $d \times d$ to $(d+1) \times d$. Thus, (3) can be rewritten as

$$E_t = \sum_{i=1}^{N_t} \mu\left(\left[T_i^l X_t, \mathbf{e}\right] H - W_i^+, \left[T_i^l X_t, \mathbf{e}\right] H - W_i^-\right). \tag{4}$$

Finally, the matrix H can be learned by minimizing the following cost function with stochastic gradient descent [44]:

$$C\left(H, T^l, L_t^l, W, L_w, \lambda\right) = E_t + \lambda \|H\|_2^2 \tag{5}$$

$$H \leftarrow \underset{H}{\operatorname{argmin}}\ C\left(H, T^l, L_t^l, W, L_w, \lambda\right) \tag{6}$$

where $\lambda \|H\|_2^2$ is a regularization term with $\lambda \in \mathbb{R}$.

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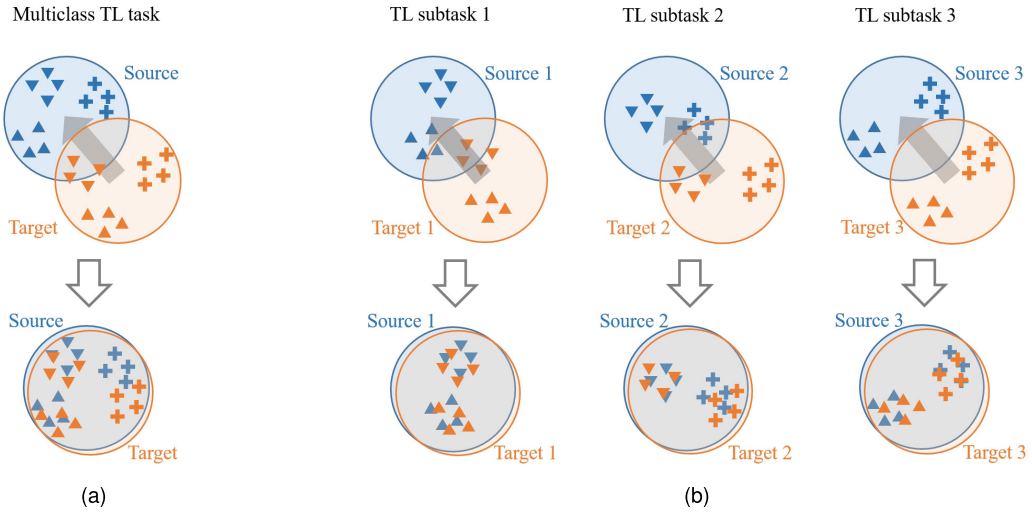


Fig. 2. Illustrations of global alignment and grouped alignment. The target and source domains share three classes, which are represented by upright triangles ▲, inverted triangles ▼, and crosshairs +, respectively. The class + has relatively small data shift than the other two classes. We also assume that any two of these classes are linearly separable. (a) Global alignment strategy ignores the fact that the data shift can vary from class to class, which might cause negative transfer locally (class +). (b) Grouped alignment strategy decomposes the 3-classes TL task into three binary subtasks. In each subtask, a linear mapping is found to align only two different classes and remove the data shift under the assumption of linear separability.

Algorithm 2 SLSA

Data : Source data S, Target data $T = \{T^l, T^u\}$, Source labels L_s, Target labels L_t^l for T_u, Subspace dimensionality d.

Result : Predicted target labels L_t^u for T^u.

$X_s \leftarrow PCA(S, d)$;
$X_t \leftarrow PCA(T, d)$;
$S_s = SX_s$;
$W, L_w \leftarrow GLVQ(S_s, L_s)$;
$T_t^l = [T^l X_t, \mathbf{e}]H$;
$H \leftarrow \text{argmin}_H C(H, T_t^l, L_t^l, W, L_w, \lambda)$;
$T_t^u = [T^u X_t, \mathbf{e}]H$;
$L_t^u \leftarrow Predict(T_t^u, W, L_w)$.

This modified algorithm is the proposed SLSA algorithm. The pseudocode is presented in Algorithm 2.

C. Grouped Subspace Linear Semantic Alignment (G-SLSA)

One problem has been raised when SLSA is applied to multiclass classification tasks. The global alignment strategy, namely, a single linear mapping for all classes, does not consider the possibility of class-level data shift. A possible consequence of such global adjustment is that some classes that are originally well aligned between the source and TARs may be incorrectly mapped, thus leading to worse classification results in the TAR.

To address this problem, we introduce a grouped alignment strategy for SLSA, thereby proposing G-SLSA. The main idea of grouped alignment is to apply a one-versus-one strategy to multiclass supervised TL. The process is as follows.

1) Obtain $K(K-1)/2$ groups from K classes by pairing two different classes into one group, i.e., in a traversal manner.
2) For each group, run SLSA with the target-domain training data from the corresponding classes.
3) Obtain the G-SLSA model that consists of $K(K-1)/2$ pairwise source-domain GLVQ classifiers and $K(K-1)/2$ linear mapping matrices.

An illustrated comparison of global alignment and grouped alignment is shown in Fig. 2. The multiclass TL task can be decomposed into several binary subtasks that can be solved efficiently using SLSA. However, another problem arises when the grouped alignment strategy is applied. As the grouped alignment reduces the number of involved classes, the size of the target training set decreases accordingly. This may not satisfy the condition for solving the linear transformation for each subtask. Therefore, to overcome the problem of limited labeled data for subtasks, we adopt the region-based growth (RG) method proposed in our previous work [45] to perform data augmentation. The initial labeled target-domain samples serve as the seeds, and RG is conducted in a seed-specific manner, in which approximately half of the neighboring pixels are selected by calculating and ranking the similarity between them and the seed.

D. Subtask Fusion for G-SLSA

For obtaining multiclass decisions from $K(K-1)/2$ binary (pairwise) classifiers, majority voting (MV) is known to be effective and efficient. However, although the voting procedure requires only pairwise decisions, it is only possible to predict single class labels in this way, whereas in many scenarios, the multiclass probabilities are desired; one example is the generation of pseudolabels [46], [47].

PWC is a technique through which a joint probability estimate for all K classes can be obtained by fusing all pairwise class probability estimates [43], [48]. However, GLVQ models do not provide probabilistic outputs. Therefore, before

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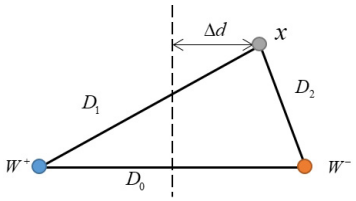


Fig. 3. Illustration of the proposed posterior probability estimation method for GLVQ. The test sample x has two nearest prototypes W^+ and W^- that belong to different classes. The dashed line passing through the midpoint of the line joining W^+ and W^- represents the separating plane. Δd indicates the distance from x to the separating plane. D_0, D_1 and D_2 indicate pairwise distances between W^+, W^- and x.

applying PWC, we must first find a suitable probability estimation for GLVQ.

1) GLVQ Posterior Probability Estimation: Wu *et al.* employed an improved implementation of Platt's method [49], [50] to estimate posterior probabilities of the binary SVM method. Each binary SVM finds an optimal separating hyperplane such that the distances from test samples to the separating hyperplane can be used to estimate the posterior probabilities by a sigmoid function. In this work, for each binary GLVQ, each test sample has two nearest prototypes W^+ and W^- that belong to positive class i and negative class j, respectively (see Fig. 3). The perpendicular plane passing through the midpoint of the line joining W^+ and W^- is the separating plane. Similarly, the distances from test samples to the separating plane (Δd) are the key to estimating binary GLVQ posterior probabilities. Instead of using Δd directly or adopting the method proposed by Platt, we propose two tricks to use Δd to obtain more accurate results.

1) Considering that the distances between different prototypes vary considerably, we divide Δd by the distance between W^+ and W^- (D_0) and obtain the relative distance ($\Delta d/D_0$).
2) The closer a test sample is to the separating plane, the closer the probability estimate is to 50%. Consequently, the binary GLVQ will have less influence on the final fusion result of this test sample. Considering a test sample is far from both W^+ and W^- likely does not belong to either of the two classes of the binary GLVQ. We want its probability estimate to be approximately 50% regardless of the distance between the test sample and the separating plane, thereby weakening the influence of this binary GLVQ on the final fusion. For this purpose, we multiply the relative distance by a factor $(D_0^2/(D_1^2+D_2^2))$ and then use the sigmoid function to estimate the binary GLVQ posterior probabilities

$$r_{ij} = \frac{1}{1+\exp\left(-\frac{\Delta d}{D_0}\frac{D_0^2}{D_1^2+D_2^2}\right)} \tag{7}$$

where D_1 and D_2 are the distances from the test sample to W^+ and W^-, respectively. When $D_1^2 + D_2^2 \gg D_0^2$, $r_{ij} \Rightarrow 50\%$; when the test sample is close to either W^+ or W^-, $(D_0^2/(D_1^2+D_2^2))$ makes relatively small adjustment to $(\Delta d/D_0)$.

The proposed GLVQ posterior probability estimation method fully exploits the information contained in a binary GLVQ model and requires no parameter optimization.

2) Class Probability Estimation via PWC: The goal of PWC is to estimate $p_i = P(y=i|x)$ by using all r_{ij}. Wu *et al.* [43] suggested that the optimal p can be obtained by solving the following optimization formulation:

$$\min_p \sum_{i=1}^{K}\sum_{j:j\neq i}(r_{ji}p_i - r_{ij}p_j)^2 \quad \text{s.t.} \sum_{i=1}^{K} p_i = 1 \tag{8}$$

$\sum_{i=1}^{K}\sum_{j:j\neq i}(r_{ji}p_i - r_{ij}p_j)^2$ can be represented in matrix from as

$$\begin{bmatrix} p_1 \\ \vdots \\ p_K \end{bmatrix}^\top \begin{bmatrix} \sum_{s:s\neq 1} r_{s1}^2 & \cdots & -r_{K1}r_{1K} \\ \vdots & \ddots & \vdots \\ -r_{1K}r_{K1} & \cdots & \sum_{s:s\neq K} r_{sK}^2 \end{bmatrix} \begin{bmatrix} p_1 \\ \vdots \\ p_K \end{bmatrix}. \tag{9}$$

Let

$$Q_{ij} = \begin{cases} \sum_{s:s\neq i} r_{si}^2, & \text{if } i=j \\ -r_{ji}r_{ij}, & \text{if } i\neq j. \end{cases} \tag{10}$$

Therefore, the objective function (8) can be rewritten as

$$\min_p 2p^\top Qp \equiv \min_p \frac{1}{2}p^T Qp. \tag{11}$$

Note that the derivative of (11) is Qp and consider that the equality constraint is $\sum_{i=1}^{k} p_i = 1$. Thus, the optimal p should satisfy the following optimality condition:

$$\begin{bmatrix} Q & \mathbf{e} \\ \mathbf{e}^T & 0 \end{bmatrix} \begin{bmatrix} p \\ b \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} \tag{12}$$

where b is the Lagrangian multiplier, $\mathbf{e}$ is a column vector of all ones, and $\mathbf{0}$ is a column vector of all zeros.

Therefore, the solution to (8) can be obtained by solving the simple linear system given in (12). For more details of the solution process, please refer to [43].

Finally, by integrating the GLVQ posterior probability estimates and the PWC scheme, we fuse the subtasks and obtain the multiclass probabilities of G-SLSA.

III. Experimental Results

A. Datasets

This section describes how three public HSI datasets are used to construct four different datasets for our experiments. These datasets exhibit different data shift between the source and TARs because the main factors causing the shift are different. The details of all datasets are listed in Tables I and II.

In the Pavia City-University of Pavia (PC-UP) dataset, the source domain is chosen to be PC (UP), and the TAR is chosen to be UP (PC). The last band of the UP dataset is removed because PC and UP share only the first 102 bands. These two

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TABLE I
DETAILS OF THE THREE PUBLIC HSI DATASETS

Dataset	Sensor	Spatial size	Spectral range	Number of bands	Spatial resolution	Location	Number of classes	Year
PC	Reflective Optics System Imaging Spectrometer	1096×92	430 to 860 nm	102	1.3 m	Pavia City	9	2002
UP	Reflective Optics System Imaging Spectrometer	610×40	430 to 860 nm	103	1.3 m	University of Pavia	9	2002
UH	Compact Airborne Spectrographic Imager	349×1905	380 to 1050 nm	144	2.5 m	University of Houston	15	2012

TABLE II
DETAILS OF THE FOUR CONSTRUCTED DATASETS

Dataset	Source domain	Target domain	Number of shared classes	Main factor inducing data shift
PC-UP	PC	UP	7	Different locations
UP-PC	UP	PC	7	Different locations
SF-CS	Shadow-free region in UH	Cloud shadow region in UH	8	Different illumination conditions
CS-SF	Cloud shadow region in UH	Shadow-free region in UH	8	Different illumination conditions

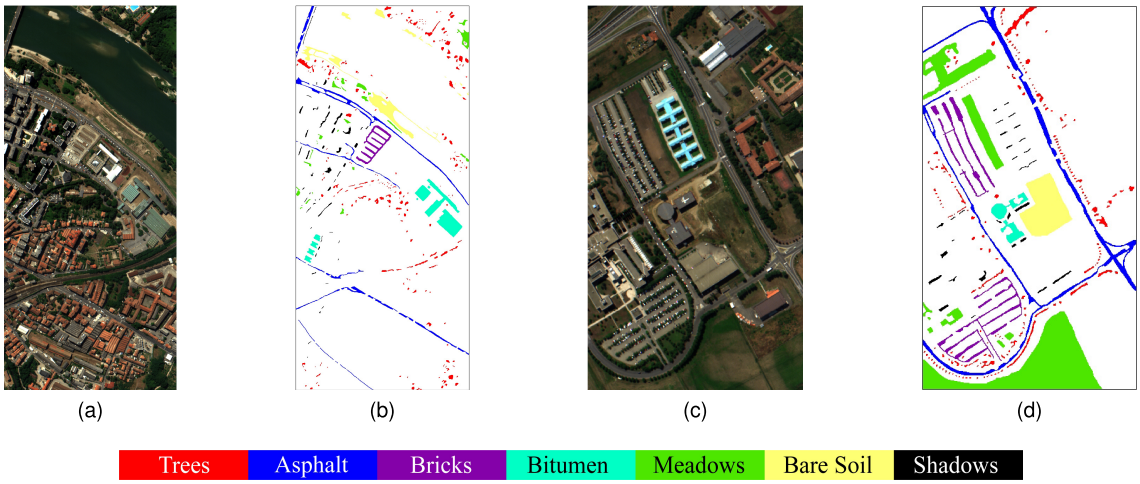


Fig. 4. PC and UP datasets. (a) False-color composite image of PC (Bands 56, 33, and 13). (b) Ground-truth map of PC. (c) False-color composite image of UP (Bands 56, 33, and 13). (d) Ground-truth map of UP.

domains each consist of nine classes, of which there are seven shared classes, and the experiments are conducted only on the shared classes. False-color composites of the PC and UP datasets and the corresponding ground-truth maps (which show only the shared classes) are shown in Fig. 4.

In the shadow-free region (SF)-CS (CS-SF) dataset, the source (target) domain is defined as the SF in the University of Houston (UH) dataset, whereas the target (source) domain consists of the manually labeled CS. The UH dataset consists of 31 classes, of which there are eight shared classes between the SF and CS, and the experiments are conducted only on the shared classes. A false-color composite of the UH dataset and the corresponding ground-truth map (which shows only the shared classes) are shown in Fig. 5.

B. Experimental Settings

1) In the proposed method, the number of prototype per class of GLVQ model is set as 1. For other free parameters, including window size and selection ratio, and subspace dimensionality, we optimize them in the following experiments.
2) For sample selection, we use different sampling strategies in the source and TARs. In the source domain, we randomly select labeled samples from the ground truth as the training set used in the experiments, i.e., 40% per class for the PC-UP and UP-PC datasets and 80% per class for the SF-CS and CS-SF datasets. In the TAR, we adopt the most common setting in few-shot learning, 5-shot classification, as a baseline; i.e., five labeled samples are assumed to be available from each class for all datasets. We also conduct 10-shot, 15-shot, and 20-shot classification in other experiments to analyze parameter settings. The remaining labeled samples are used as test samples to evaluate the classification performance.
3) For performance comparisons, we select three traditional supervised TL methods, including A-SVM [38], LSTL [40] and NFNalign [39]; and three state-of-the-art deep TL methods, including classification and

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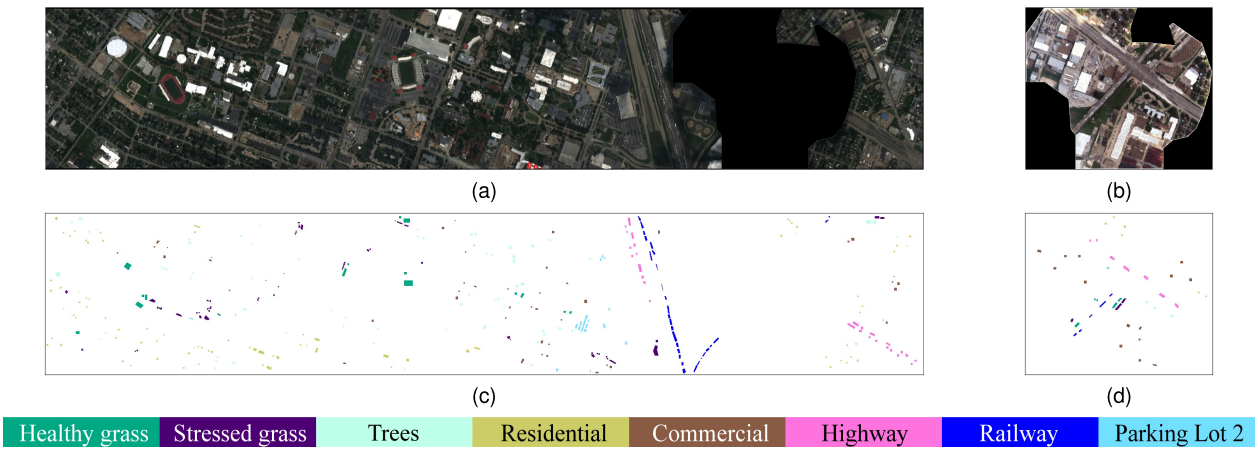


Fig. 5. UH dataset. (a) False-color composite image of SF (Bands 59, 40, and 23). (b) False-color composite image of CS (Bands 59, 40, and 23; after contrast stretching). (c) Ground-truth map of SF. (d) Ground-truth map of CS.

contrastive semantic alignment (CCSA) [34], minimax entropy (MME) [51], and deep cross-domain few-shot learning (DCFSL) [36]. Meanwhile, SA-SVM (the SA algorithm [20] using a linear SVM as its classifier), TAR-SVM (a linear SVM trained on target-domain data only) is chosen as the baseline approach. For A-SVM, we set the cost factor to $C = 9$ and use the radial basis function (RBF) kernel $\exp(-\gamma \|x_i - x_j\|^2)$ with $\gamma = 0.1$ to obtain optimal results. For LSTL, the number of prototype per class of GLVQ model is set as 1. For NFNalign, we use the spectral pixel correspondence approach provide in [39] to accommodate our disjoint datasets; the reference spectra B for NFN are defined from the highest density point per class of the source-domain training set which can be obtained by mean shift algorithm. For CCSA, we follow the basic Siamese architecture presented in [34] and set the kernel size of all convolutional blocks as 1×3. The number of CCSA training epochs is set to 100. For MME, we follow the basic model architecture presented in [51] and change the backbone to 1D_Convolutional Neural Network (CNN) [52]. The number of MME iterations is set to 10 000. For DCFSL, the kernel size of all 3-D convolutional blocks is set as $1 \times 1 \times 3$ for a fair comparison, and the number of episodes of DCFSL is set to 20 000. Adam is chosen as the optimizer for CCSA, MME and DCFSL, the batch size is set to 128, and the learning rate is set to 0.001 to obtain optimal results.

4) To ensure fair comparisons, for all methods except SA-SVM, both the initial training samples and the augmented samples are used as target-domain training set.
5) All implementations are carried out using MATLAB, Pyotrch, and Keras (using the TensorFlow backend) on a Windows LTSC desktop equipped with a GeForce RTX 2070 SUPER graphics processing unit (GPU), an Intel i7-9700K CPU, and 64 GB of RAM. The overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), training time (s), and test time (s) are adopted to evaluate the classification performance. We run the experiment 20 times with random initial training samples.

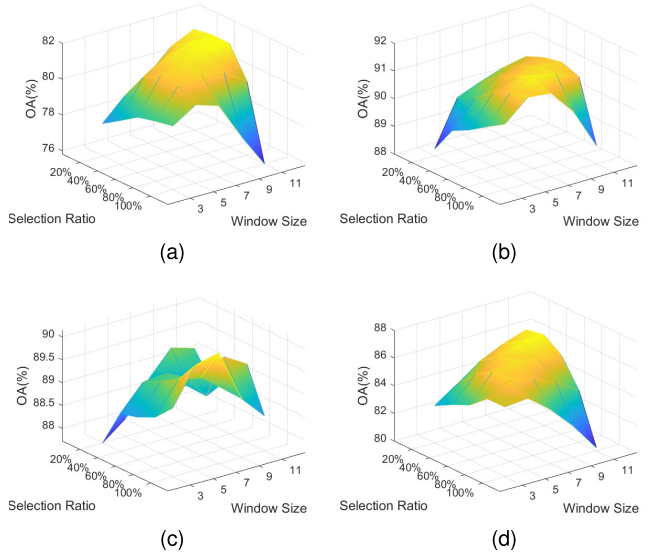


Fig. 6. RG data augmentation to reduce sample selection bias. (a) PC-UP. (b) UP-PC. (c) SF-CS. (d) CS-SF.

C. Analysis of Data Augmentation

In the first experiment, we analyze the effect of RG data augmentation on classification accuracy. First, we investigate the suitable RG parameters (window size (w) and selection ratio (r)) setting by assigning $w \in [3, 5, \ldots, 11]$ and $r \in [20\%, 40\%, \ldots, 100\%]$. As shown in Fig. 6(a), (b), and (d), the 3-D plots illustrate a ridge-like trend of OAs as w and r change. The classification accuracy increases as w increases and decreases as r increases, which is because: 1) a larger window size will collect more target-domain information however and 2) a larger selection ratio will produce more incorrectly labeled augmentation samples. For the SF-CS dataset [see Fig. 6(c)], a large w with a small r will not produce an OA

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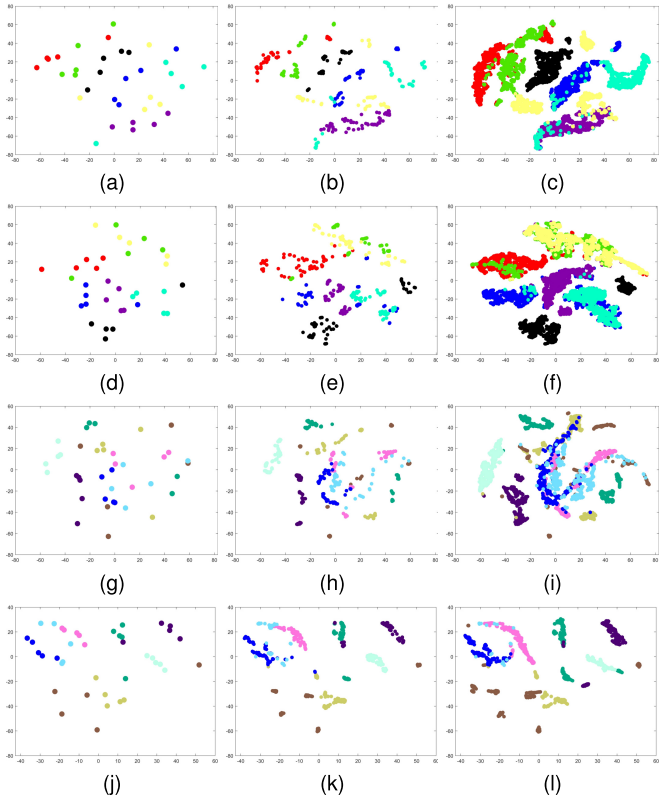


Fig. 7. TAR t-SNE visualization for different datasets of seeds (Seeds, left plots), augmented samples (Augmentation, middle plots), and ground truth (Ground Truth, right plots). The x-axis and y-axis correspond to two dimensions of the 2-D t-SNE visualization results. RG data augmentation not only increases the training set but also reduces sample selection bias by expanding the distribution of the training set. (a) PC: Seeds. (b) PC: Augmentation. (c) PC: Ground Truth. (d) UP: Seeds. (e) UP: Augmentation. (f) UP: Ground Truth. (g) SF: Seeds. (h) SF: Augmentation. (i) SF: Ground Truth. (j) CS: Seeds. (k) CS: Augmentation. (l) CS: Ground Truth.

as good as a small w with a large r. This is because the TAR in the SF-CS dataset (CS) has a low contrast, which affects the discriminability of the similarity in RG. Consequently, the RG will not work well for selecting reliable augmentation samples, and the incorrectly labeled augmentation samples increase, thereby degrading the classification accuracy. Therefore, for robustness, we set the parameter of RG to $w = 5$ and $r = 60\%$ in the following experiments for all datasets.

Second, we analyze the effect of RG data augmentation on reducing sample selection bias. According to the mentioned parameter setting, almost 75 augmented samples (including the seeds) can be obtained for each class. To demonstrate the effect of RG data augmentation, we apply t-distributed stochastic neighbor embedding (t-SNE) to visualize the features of the seeds, augmented samples, and ground truth. As shown in Fig. 7, a limited number of seeds (left plots) results in sample selection bias, which leads to skewed estimates of the true underlying distribution of the classes (right plots). As seen from the middle plots, the RG strategy for data augmentation not only increases the size of the training set but also expands its distribution. As a result, the sample selection bias can be considerably reduced under the few-shot setting.

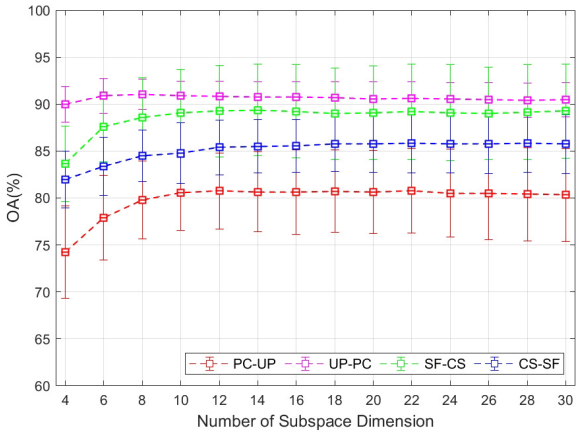


Fig. 8. Influence of the subspace dimensionality d on G-SLSA.

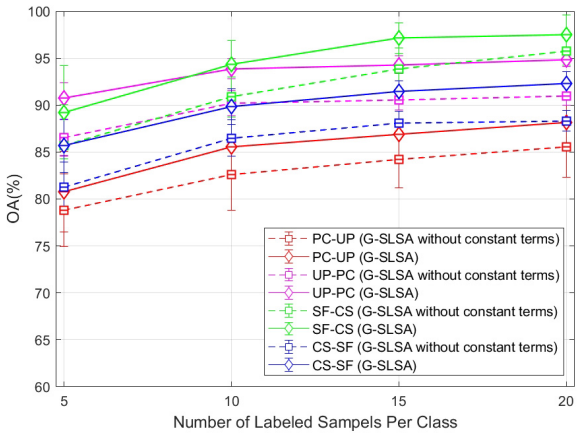


Fig. 9. Influence of constant terms on G-SLSA.

D. Analysis of the Number of Subspace Dimensions

In the second experiment, we analyze the influence of the number of subspace dimensions, which affects the size of the linear mapping matrix and, in turn, affects the alignment performance. Fig. 8 shows the OA of different datasets with different subspace dimensionalities. Generally, there is a trend of an initial rapid increase followed by a slow decrease in the OA. A subspace dimensionality that is too low can cause the solution to degrade because of the loss of information in the process of dimension reduction, whereas a higher subspace dimensionality will cause the alignment performance to degrade. To balance the influences of the information loss and the size of the mapping matrix, as mentioned above, the number of subspace dimensions is set to 15 in the following experiments for all datasets.

Fig. 8 also shows the large variances of the results. This may be explained by the fact that: 1) the experiments are conducted under the few-shot setting and 2) the proposed method only uses spectral feature.

E. Analysis of the Introduction of Constant Terms

In the third experiment, we analyze the effect of the constant terms on the OA (see Fig. 9). Generally, introducing

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TABLE III
CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR THE PC-UP DATASET

Class	SA-SVM [20]	TAR-SVM	A-SVM [38]	LSTL [40]	NFNalign [39]	CCSA [34]	MME [51]	DCFSL [36]	G-SLSA
Trees	80.90±2.15	89.38±7.36	**93.84±3.24**	77.35±7.98	67.89±26.17	90.56±5.03	88.07±8.07	90.61±9.00	87.39±8.57
Asphalt	**90.89±1.37**	79.56±8.13	79.41±6.80	69.08±6.05	82.44±9.11	77.40±6.66	73.63±8.02	81.51±5.75	76.20±6.56
Bricks	**95.77±2.07**	92.13±6.69	88.32±6.34	77.65±8.47	89.39±6.97	93.51±4.68	94.81±2.23	94.24±3.99	95.30±2.71
Bitumen	13.92±7.78	92.30±6.74	89.16±6.14	81.20±8.47	63.00±35.86	94.60±4.65	92.85±5.25	94.15±4.31	**95.94±2.31**
Meadows	36.71±1.16	69.69±11.93	44.61±11.45	66.81±9.30	53.63±11.64	63.98±12.87	68.93±9.79	62.97±14.09	**77.34±10.61**
Bare Soil	81.28±0.87	70.71±13.49	57.88±10.57	62.54±7.76	72.56±12.22	76.84±13.30	**81.76±10.15**	72.65±10.87	77.46±12.47
Shadows	90.88±3.41	97.55±5.37	**99.97±0.05**	99.68±0.38	97.96±3.29	98.84±1.71	98.50±2.04	99.55±0.60	99.56±0.62
OA(%)	61.04±0.55	76.55±5.27	62.93±5.60	69.76±4.55	66.74±5.28	74.58±4.66	76.79±3.77	74.34±5.76	**80.78±4.28**
AA(%)	70.05±1.26	84.48±2.56	79.03±2.55	76.33±2.37	75.27±5.98	85.10±2.27	85.51±2.08	85.10±2.46	**87.03±1.79**
κ	0.519±0.006	0.692±0.060	0.543±0.059	0.604±0.049	0.570±0.058	0.671±0.051	0.695±0.044	0.669±0.064	**0.744±0.051**

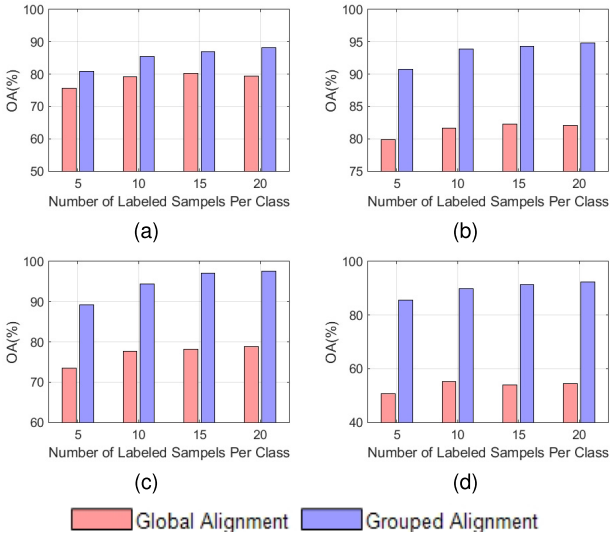


Fig. 10. Influence of the grouped alignment strategy on SLSA. (a) PC-UP. (b) UP-PC. (c) SF-CS. (d) CS-SF.

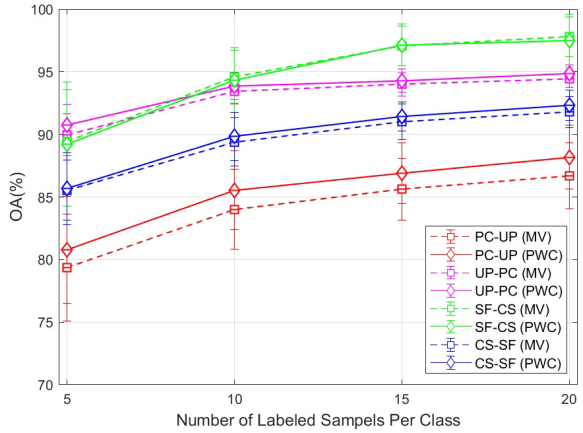


Fig. 11. G-SLSA using different subtask fusion methods.

constant terms will lead to higher accuracy for different datasets and different numbers of labeled samples. For the UP-PC dataset, the OA achieved by G-SLSA is 3.70%–4.23% higher than that achieved by G-SLSA without the constant terms. For the SF-CS dataset, the improvement in OA is as significant as it is for UP-PC (i.e., 3.32%–4.38%). Relatively small improvements can also be observed for the PC-UP and CS-SF datasets, which are 1.97%–2.95% and 1.78%–3.53%, respectively.

F. Analysis of the Application of the Grouped Alignment Strategy

In the fourth experiment, we analyze the effect of the grouped alignment strategy on the OA. As shown in Fig. 10, SLSA that employs the grouped alignment strategy (G-SLSA) exhibits better classification performance compared to employing the global alignment strategy (original SLSA). For instance, G-SLSA achieves an OA of 80.78% for the PC-UP dataset with five labeled samples per class, whereas SLSA achieves an OA of 76.16%. As for different initial training sample numbers, the grouped alignment also considerably improves OAs. For the other three datasets, the same conclusion holds.

G. Analysis of the Subtask Fusion Strategy in G-SLSA

In the fifth experiment, we compare the subtask fusion performance of MV and PWC. As shown in Fig. 11, PWC is found to outperform MV in most of the cases for different datasets. For instance, the OA obtained by G-SLSA using PWC is 1.29%–1.53% higher than using MV for the PC-UP dataset. Clear improvements can also be found for UP-PC and CS-SF datasets. For the SF-CS datasets, MV provides competitive performance.

H. Classification Results

In the sixth experiment, we provide a comprehensive comparison of different methods based on limited target-domain training samples (five initial training samples + approximately 70 augmented samples per class), and enough source-domain training samples, i.e., 40% per class for the PC-UP and UP-PC datasets, and 80% per class for the SF-CS and CS-SF datasets. Note that the classification maps we provide in Figs. 12–15 are the one that obtains the highest OA among 20 independent runs for each method.

1) For the PC-UP dataset, which is the most challenging dataset, G-SLSA obtains the highest OA of $80.78 \pm 4.28\%$, which is 4.23% higher than TAR-SVM, 11.02%–17.85% higher than the three traditional methods, and 3.99%–6.44% higher than the three DL methods in Table III. For AA and κ, G-SLSA also produces significant improvements compared with other

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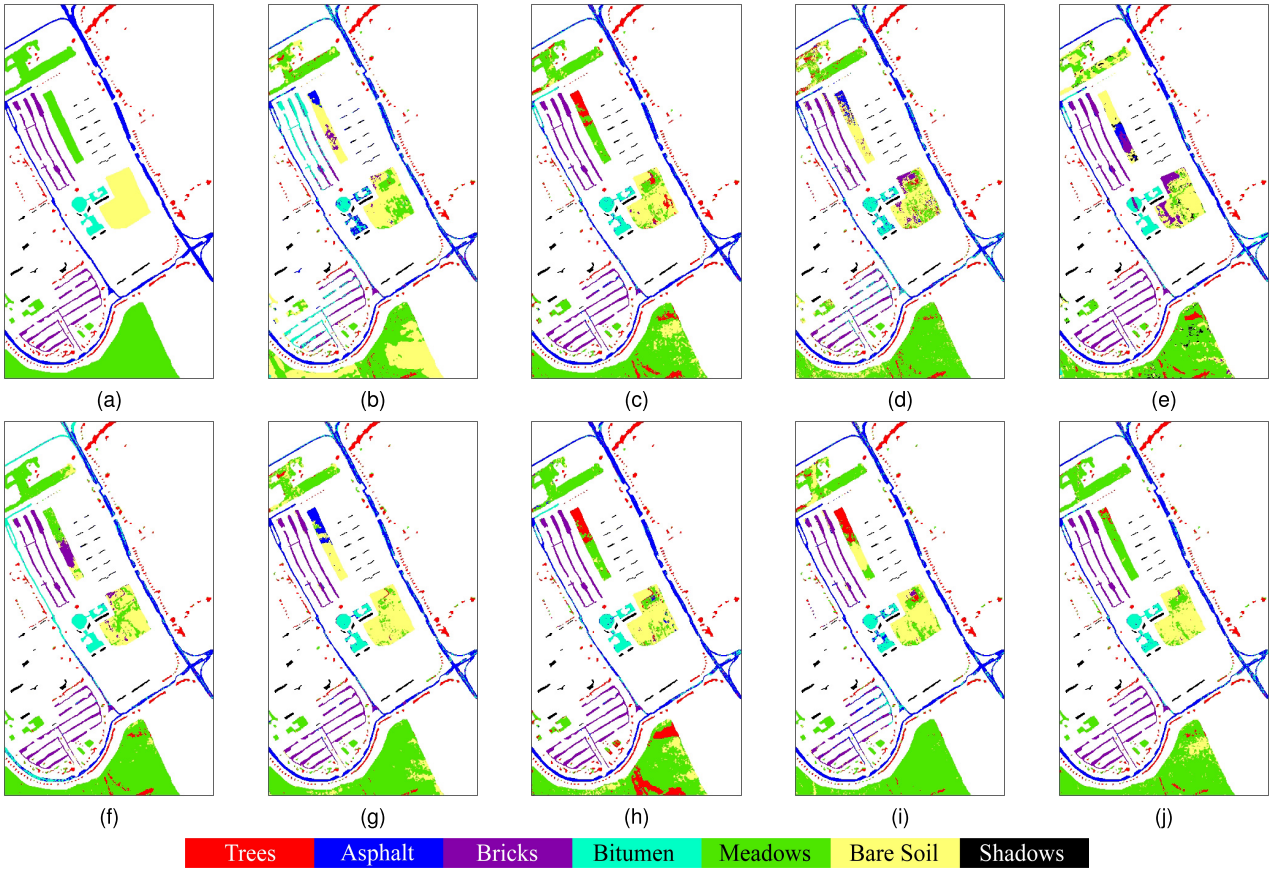


Fig. 12. Classification maps for the PC-UP dataset with five training samples per class and augmented samples. (a) Ground truth. (b) SA-SVM (62.59%). (c) TAR-SVM (83.10%). (d) A-SVM (74.02%). (e) LSTL (75.82%). (f) NFNalign (74.72%). (g) CCSA (83.26%). (h) MME (83.01%). (i) DCFSL (82.95%). (j) G-SLSA (88.24%).

TABLE IV
CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR THE UP-PC DATASET

Class	SA-SVM [20]	TAR-SVM	A-SVM [38]	LSTL [40]	NFNalign [39]	CCSA [34]	MME [51]	DCFSL [36]	G-SLSA
Trees	84.49±2.19	90.71±5.01	83.76±7.99	79.02±9.01	91.27±6.30	92.02±4.47	**96.44±1.59**	90.57±6.86	92.57±3.34
Asphalt	**99.91±0.01**	96.11±3.06	94.16±4.02	94.09±1.82	95.40±4.23	95.14±4.45	97.52±5.01	97.07±1.97	96.91±3.14
Bricks	80.82±1.95	91.78±9.09	89.28±7.69	73.30±8.18	93.54±6.02	**96.28±2.97**	87.80±3.81	92.53±6.35	92.50±5.13
Bitumen	0.01±0.01	84.08±7.98	73.11±12.30	69.24±11.33	77.45±8.06	85.67±6.33	83.57±3.94	85.77±5.83	**86.70±4.63**
Meadows	**93.38±0.54**	85.47±9.04	89.41±7.82	72.78±7.00	37.85±25.69	89.04±5.77	83.64±5.20	89.27±6.66	86.75±9.41
Bare Soil	84.58±1.24	**86.54±7.31**	70.84±9.44	62.97±11.02	83.45±12.76	83.06±10.25	85.17±4.88	86.12±9.19	84.49±9.31
Shadows	99.99±0.03	99.89±0.18	98.64±3.07	99.93±0.09	**100.00±0.00**	99.97±0.05	**100.00±0.00**	99.99±0.02	99.98±0.04
OA(%)	71.78±0.48	89.92±1.63	83.11±3.68	77.68±2.64	84.10±3.59	90.21±1.81	90.54±1.43	90.74±2.27	**90.75±1.64**
AA(%)	77.60±0.43	90.65±1.65	85.60±3.23	78.76±1.73	82.71±4.12	91.60±1.50	90.59±1.17	**91.62±1.88**	91.41±1.39
κ	0.660±0.006	0.879±0.020	0.799±0.043	0.734±0.031	0.809±0.043	0.883±0.021	0.886±0.017	**0.889±0.027**	**0.889±0.020**

methods. For the class-specific accuracies, G-SLSA obtains the highest accuracy of the Bitumen class and Meadows class. For the Meadows class, which is an especially challenging class for accurate classification, G-SLSA obtains 77.34% accuracy which is 7.0% higher than the second best method. The classification maps in Fig. 12 also demonstrate that G-SLSA reaches a better result for the Meadows class. The high performance of G-SLSA on this difficult dataset fully validates its superiority.

2) For the UP-PC dataset, TAR-SVM, CCSA, MME, and DCFSL yield competitive results in Table IV. However, G-SLSA still achieves the highest OA of 90.75±1.64%, which is 0.01%–0.83% higher than TAR-SVM and the DL methods, 6.65%–13.07% higher than the traditional methods. For the class-specific accuracies, the proposed method obtains the highest accuracy in the Bitumen class (i.e., 86.7%), which is difficult to be accurately classify in this scene. TAR-SVM, CCSA, and MME exhibit higher accuracy for the Bare soil class, Bricks class, and Trees class, respectively. These results demonstrate that simple classifier with data augmentation in the TAR may achieve performance as good as that of state-of-the-art TL methods.
3) For the SF-CS dataset, G-SLSA obtains the highest OA of 89.22 ± 4.95%, which is 2.56% higher than

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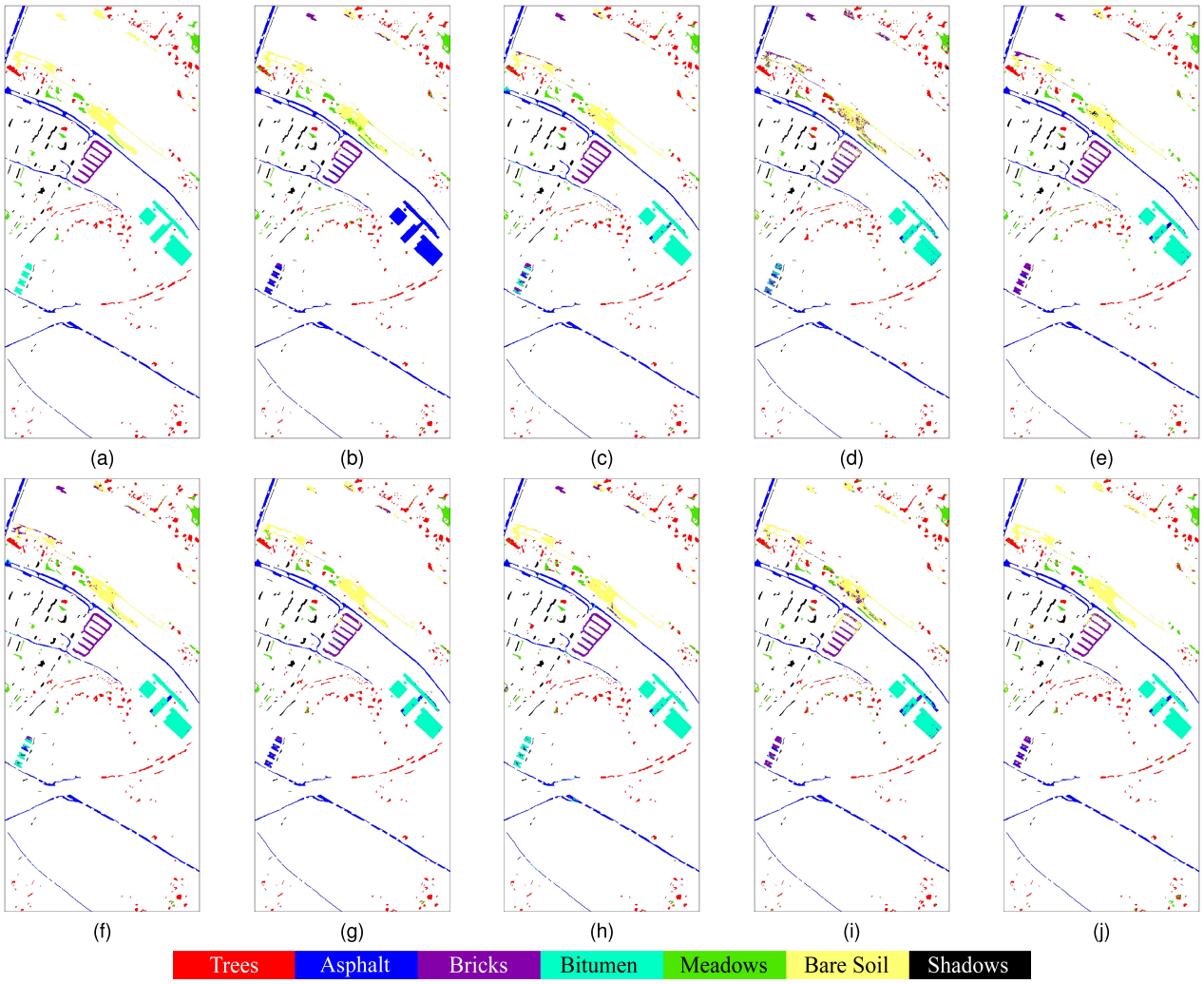


Fig. 13. Classification maps for the UP-PC dataset with five training samples per class and augmented samples. (a) Ground truth. (b) SA-SVM (72.57%). (c) TAR-SVM (92.65%). (d) LSTL (82.91%). (e) A-SVM (88.11%). (f) NFNalign (89.46%). (g) CCSA (93.28%). (h) MME (92.33%). (i) DCFSL (90.30%). (j) G-SLSA (92.91%).

TABLE V
CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR THE SF-CS DATASET

Class	SA-SVM [20]	TAR-SVM	A-SVM [38]	LSTL [40]	NFNalign [39]	CCSA [34]	MME [51]	DCFSL [36]	G-SLSA
Healthy grass	0.00±0.00	84.83±7.68	83.64±9.33	78.79±5.82	87.51±8.63	86.39±9.17	**88.64±6.25**	86.91±9.55	88.55±8.30
Stressed grass	52.25±25.80	96.05±6.84	**97.22±3.03**	90.36±5.34	93.56±8.56	94.35±7.03	91.70±9.30	92.52±9.14	95.07±6.03
Trees	60.31±30.37	99.77±1.03	99.69±0.95	88.08±7.07	99.46±1.15	99.77±1.03	99.23±2.09	99.92±0.34	**100.00±0.00**
Residential	**100.00±0.00**	85.47±10.14	88.80±6.62	66.27±11.99	86.33±7.81	86.53±9.59	89.00±8.27	84.13±10.03	84.27±8.97
Commercial	34.78±11.74	77.32±11.46	**86.31±11.73**	76.64±12.86	81.42±11.31	84.89±11.63	81.67±18.48	82.28±13.17	81.69±12.45
Highway	68.94±2.51	89.56±9.93	86.71±10.94	66.37±10.90	84.64±8.53	91.01±9.98	91.64±7.94	87.87±12.04	**92.94±10.82**
Railway	10.22±1.55	94.11±10.14	88.15±11.98	58.26±8.95	77.00±23.29	91.56±9.84	81.41±18.71	88.74±12.67	**95.33±5.78**
Parking Lot 2	0.33±1.49	97.33±6.63	93.00±4.58	66.33±13.06	38.00±41.33	**98.67±2.74**	63.33±23.55	96.00±7.12	94.67±13.61
OA(%)	42.37±4.67	86.66±5.23	88.32±5.53	74.08±3.40	84.53±4.30	89.20±4.66	87.04±6.19	87.07±5.91	**89.22±4.95**
AA(%)	40.85±4.94	90.56±3.62	90.44±3.39	73.89±2.52	80.99±6.34	**91.65±3.26**	85.83±3.56	89.80±3.43	91.57±3.56
κ	0.35±0.05	0.839±0.061	0.859±0.064	0.684±0.039	0.810±0.053	0.868±0.055	0.844±0.070	0.843±0.069	**0.869±0.058**

TAR-SVM, 0.02%–2.18% higher than the DL methods, and 0.9%–15.14% higher than the traditional methods in Table V. CCSA obtains the highest AA of 91.65±3.26%, which is 0.08% higher than that of G-SLSA. For κ, G-SLSA also produces higher results than other methods. For the class-specific accuracies, G-SLSA obtains better classification results for a total of three classes. Specifically, G-SLSA obtains 100% accuracy for the Trees class. Generally, for this dataset, all supervised TL methods except LSTL and NFNalign produce better results than TAR-SVM. This phenomenon indicates that these supervised TL methods successfully transfer knowledge from the source domain (SF) to the TAR (CS). TAR-SVM cannot achieve equal classification performance without the information from the source domain.

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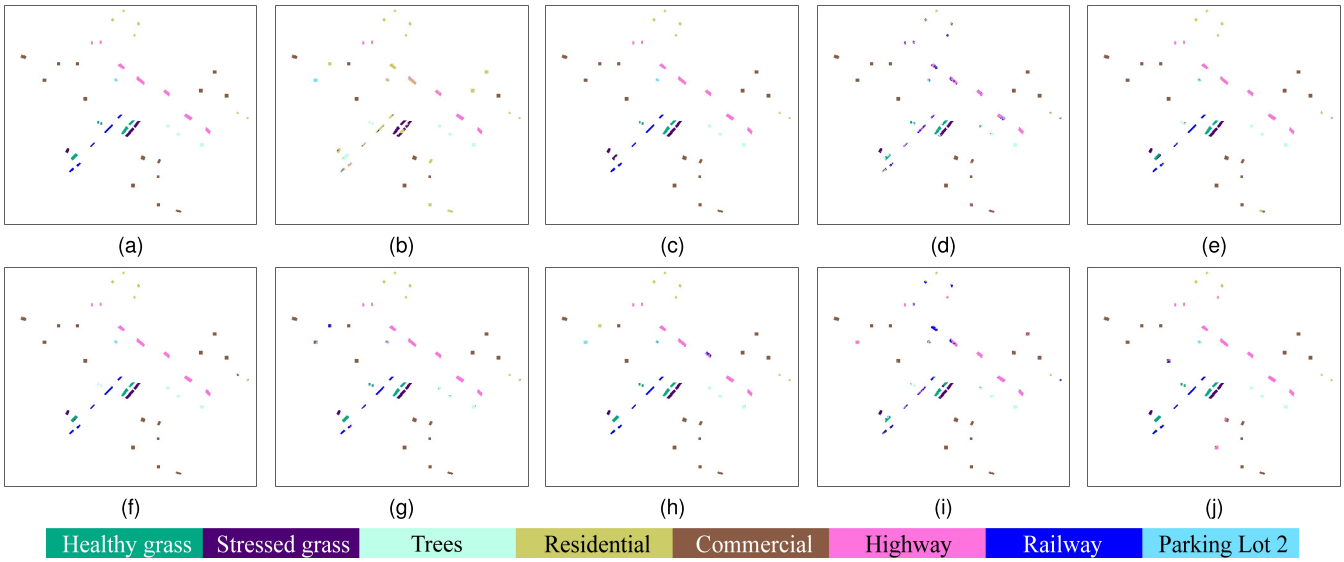


Fig. 14. Classification maps for the SF-CS dataset with five training samples per class and augmented samples. (a) Ground truth. (b) SA-SVM (50.48%). (c) TAR-SVM (94.79%). (d) LSTL (80.18%). (e) A-SVM (93.76%). (f) NFNalign (93.03%). (g) CCSA (96.04%). (h) MME (95.67%). (i) DCFSL (94.50%). (j) G-SLSA (97.21%).

TABLE VI
CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR THE CS-SF DATASET

Class	SA-SVM [20]	TAR-SVM	A-SVM [38]	LSTL [40]	NFNalign [39]	CCSA [34]	MME [51]	DCFSL [36]	G-SLSA
Healthy grass	**100.00±0.00**	97.08±2.81	94.48±5.43	59.58±7.25	98.82±0.89	98.03±2.73	98.38±1.44	98.24±1.29	98.83±1.05
Stressed grass	52.75±18.39	**96.87±5.78**	94.95±6.59	45.05±9.39	94.83±4.74	93.05±8.10	88.71±9.19	93.72±7.37	93.61±6.03
Trees	0.98±0.13	96.49±3.85	87.89±8.28	56.63±7.39	91.69±4.12	94.54±2.72	94.97±8.34	92.43±5.68	**97.75±2.32**
Residential	0.33±0.12	**82.27±12.09**	72.08±14.64	52.06±12.18	67.33±15.33	75.37±12.78	72.22±10.43	78.20±12.05	77.94±14.09
Commercial	**70.27±2.08**	54.26±11.43	54.43±5.90	59.71±10.09	32.51±20.94	56.27±11.10	55.81±13.04	47.48±12.99	48.75±7.49
Highway	0.00±0.00	80.05±14.27	74.12±17.20	44.41±10.54	76.21±15.43	79.35±16.44	80.53±15.38	81.01±14.76	**89.54±9.67**
Railway	8.39±3.12	81.18±14.89	64.92±12.54	57.03±11.89	63.53±12.49	74.40±11.41	73.54±13.47	77.07±14.05	**87.19±14.87**
Parking Lot 2	0.00±0.00	80.86±7.72	79.54±4.66	18.03±5.57	2.98±6.09	74.52±9.19	70.47±10.84	**83.73±6.32**	78.83±4.95
OA(%)	29.91±2.59	85.10±2.67	78.59±3.32	51.46±3.14	72.39±3.14	82.16±3.18	80.91±2.85	82.57±2.86	**85.66±2.86**
AA(%)	29.09±2.28	83.63±2.74	77.80±3.22	49.06±3.00	65.99±3.34	80.69±3.05	79.33±3.20	81.48±2.84	**84.06±2.56**
κ	0.199±0.030	0.829±0.031	0.755±0.038	0.437±0.04	0.680±0.037	0.795±0.036	0.780±0.033	0.799±0.033	**0.835±0.033**

4) For the CS-SF dataset, G-SLSA obtains the highest OA of 85.66 ± 2.86%, which is 0.56% higher than TAR-SVM, 3.09%–4.75% higher than the three DL methods, and 7.07% higher than the highest one among the three traditional methods in Table VI. For AA and κ, G-SLSA produces the highest results. For the class-specific accuracies, G-SLSA obtains significant improvement in the classification performance of the Highway class and the Railway class. Similar to the PC-UP dataset, DL methods again obtains relatively worse results. A possible reason for this phenomenon is as follows. When the target-domain data are difficult and lacking in labels, DL methods will fail to maintain high performance, whereas G-SLSA can still be effective because of its linearity and the simplification of data shift.

IV. DISCUSSION

A. Space Complexity Analysis

For the memory burden, the space complexity of G-SLSA can be formulated as

$$O(K^2P). \tag{13}$$

This would be a potential limitation of G-SLSA because the grouped alignment strategy requires more memory to allocate subtask models on the order of K^2.

B. Computational Complexity Analysis

Let us suppose that all K classes have nearly equal training samples. The computational complexity of G-SLSA in the training phase can be formulated as

$$O\left(\frac{N_s^3}{K}+\frac{N_t^3}{K}+N_sKPI_s+N_tKPI_t\right) \tag{14}$$

where N_s is the number of source-domain training samples, N_t is the number of target-domain training samples, K is the number of classes, P is the number of prototypes per class in source-domain GLVQ model, I_s is the number of iterations for training the GLVQ model, I_t is the number of iterations for learning the linear mapping.

Accordingly, the computational complexity of G-SLSA in the test phase can be formulated as

$$O(NK^2P) \tag{15}$$

where N is the number of target-domain test samples.

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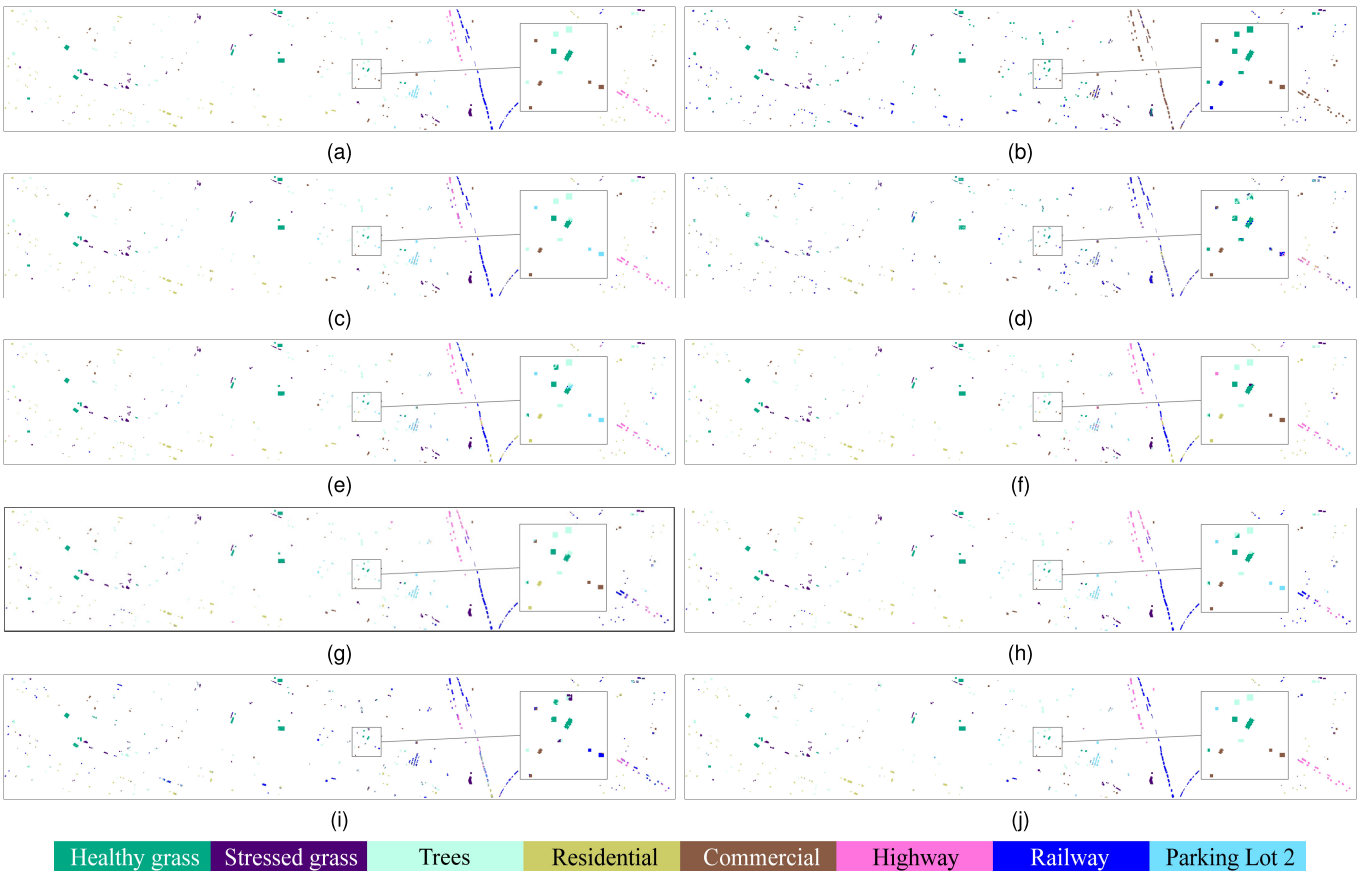


Fig. 15. Classification maps for the CS-SF dataset with five training samples per class and augmented samples. (a) Ground truth. (b) SA-SVM (34.78%). (c) TAR-SVM (89.55%). (d) LSTL (55.98%). (e) A-SVM (84.08%). (f) NFNalign (77.07%). (g) CCSA (86.31%). (h) MME (85.26%). (i) DCFSL (87.51%). (j) G-SLSA (90.02%).

TABLE VII
RUNNING TIMES OF DIFFERENT METHODS

Dataset	Time	SA-SVM [20]	TAR-SVM	A-SVM [38]	LSTL [40]	NFNalign [39]	CCSA [34]	MME [51]	DCFSL [36]	G-SLSA
PC-UP	Training Time(s)	2.99	**0.07**	2.81	29.06	14.03	69.82	384.12	510.58	1.67
	Test Time(s)	0.09	0.69	0.80	**0.10**	6.91	1.79	6.21	1.92	1.38
UP-PC	Training Time(s)	4.61	**0.08**	12.83	36.54	17.00	68.14	440.93	512.40	1.93
	Test Time(s)	**0.08**	0.62	0.70	0.09	6.30	1.37	5.13	1.88	1.24
SF-CS	Training Time(s)	4.24	**0.09**	0.87	0.57	6.97	133.43	355.42	451.02	0.90
	Test Time(s)	**0.01**	0.06	0.06	**0.01**	0.15	0.49	0.14	0.08	0.05
CS-SF	Training Time(s)	3.98	**0.09**	0.40	2.54	2.80	140.09	340.26	467.28	0.50
	Test Time(s)	0.03	0.25	0.27	**0.03**	0.15	0.61	1.91	0.42	0.29

We report the running time by different methods in Table VII. In the training phase, the running time of the proposed method needs is approximately equal to the time needed by the other traditional ML methods but much less than the time needed by DL methods. Compared with the SLSA, the G-SLSA has a remarkable improvement in the training time. This is mainly because the optimization of the GLVQ cost function (1) will converge much faster in the binary setting than in the multiclass setting. Considering $N_s \gg N_t$, fewer iterations save considerable time during source-domain GLVQ model training in G-SLSA. In the test phase, G-SLSA requires a much longer time than the others because it makes $K(K-1)/2$ predictions for each test sample.

C. Effects of the Number of Prototypes Per Class on G-SLSA

We note that a GLVQ model that considers more prototypes tends to yield better results for source-domain classification tasks [Fig. 16(a)]. However, this efficacy cannot be "transferred" to the proposed method. As shown in Fig. 16(b), as the number of prototypes per class increases, the opposite trend is seen for the target-domain test OAs. To explore the reason for the deterioration of the target-domain test OAs, we introduce a statistic for evaluating the validity of the prototypes, namely, the average valid prototype ratio (AVPR), which is defined as

$$\text{AVPR} = \frac{1}{K(K-1)/2} \sum_{k=1}^{K(K-1)/2} \frac{\text{Num}(V_k)}{\text{Num}(P_k)} \tag{16}$$

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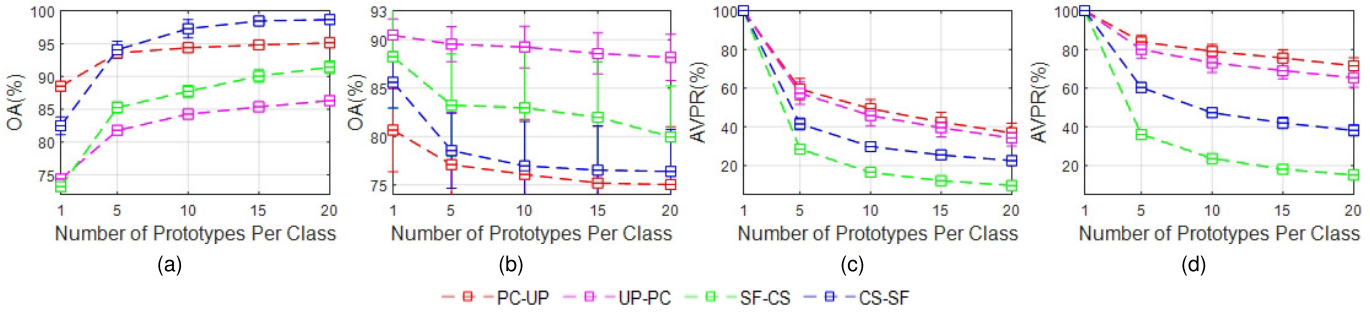


Fig. 16. Influence of the number of prototypes per class on G-SLSA. (a) Source-domain test OAs. (b) TAR test OAs. (c) Target-domain training AVPRs. (d) TAR test AVPRs.

where P denotes all prototypes in a pairwise GLVQ classifier and the valid prototypes (V) are defined as

$$V = \{v \in P | \exists x \in X : \text{class}(x) = \text{class}(v) \quad \text{s.t. } \|v - x\|_2^2 = \min\{\|p_j - x\|_2^2 | p_j \in P\}\} \tag{17}$$

where X denotes the transformed target-domain data, $\text{class}(x)$ is the ground-truth class of x, and $\text{class}(v)$ is the class of v.

According to (16), each valid prototype has at least one target test sample of the same class belonging to it. Therefore, we can use the AVPR to evaluate and compare the validity of prototypes. As depicted in Fig. 16(c) and (d), in both the training and test phases, the validity of the prototypes shows no improvement but rather deteriorates with an increasing number of prototypes. Low training AVPRs suggest that the target-domain training data are aligned with only a few valid prototypes. Consequently, it is not possible to control how the TAR will align with the remaining invalid prototypes in the test phase. Although in Fig. 16(d), some of the invalid prototypes become valid and likely increase the target-domain generalization performance, the decreasing trends of the target-domain test AVPRs and OAs demonstrate that an excess of prototypes leads to a more severely negative transfer and degrades the performance.

V. Conclusion

In this article, we present a novel linear TL method, namely, G-SLSA, for HSI classification. This article makes the following arguments:

1) The data shift between two HSIs can be reduced by learning a linear transformation function that semantically aligns the source-domain subspace with the target-domain subspace.
2) The experimental results show that the proposed grouped alignment strategy is extremely beneficial for SLSA in multiclass scenarios.
3) The experiments also demonstrate the effectiveness of the proposed G-SLSA multiclass probability estimation method and indirectly demonstrate the accuracy and robustness of the proposed GLVQ posterior probability estimation method.

Several changes could lead to further improvements. In our future work, we plan to improve the alignment performance by semisupervised techniques, which may help to activate more prototypes for the target-domain classification task. We also plan to expand the proposed method to address the class-incremental domain adaptation problem, in which the TAR is allowed to have novel (target-private) classes not shared with the source domain.

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